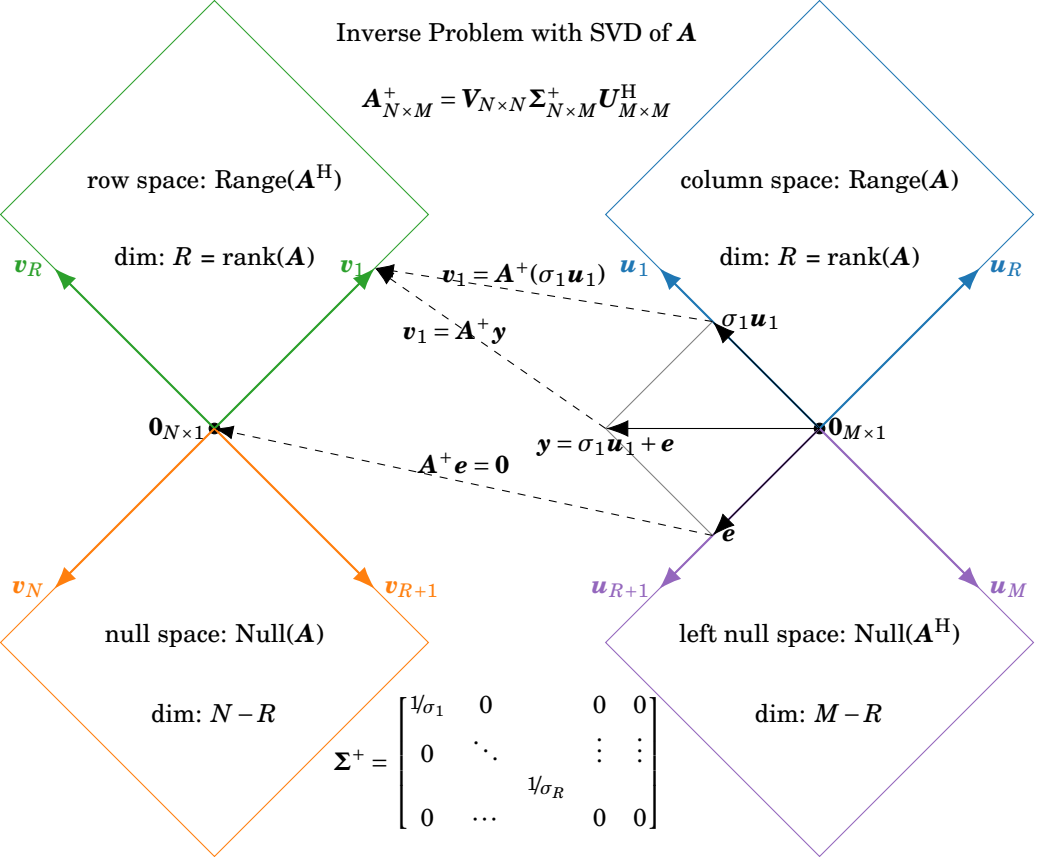
Inverse Problem with SVD of $\boldsymbol{A}$
$\boldsymbol{A}^{+}_{N\times M} = \boldsymbol{V}_{N\times N}\boldsymbol{\Sigma}^{+}_{N\times M}\boldsymbol{U}^{\mathrm{H}}_{M\times M}$
row space: Range($\boldsymbol{A}^{\mathrm{H}}$)
dim: R = rank($\boldsymbol{A}$)
column space: Range($\boldsymbol{A}$)
dim: R = rank($\boldsymbol{A}$)
$\boldsymbol{v}_R$
$\boldsymbol{v}_1$
$\boldsymbol{u}_1$
$\boldsymbol{u}_R$
$\boldsymbol{v}_1 = \boldsymbol{A}^{+}(\sigma_1\boldsymbol{u}_1)$
$\boldsymbol{v}_1 = \boldsymbol{A}^{+}\boldsymbol{y}$
$\sigma_1\boldsymbol{u}_1$
$\mathbf{0}_{N\times 1}$
$\mathbf{0}_{M\times 1}$
$\boldsymbol{y} = \sigma_1\boldsymbol{u}_1 + \boldsymbol{e}$
$\boldsymbol{A}^{+}\boldsymbol{e} = \mathbf{0}$
$\boldsymbol{e}$
$\boldsymbol{v}_N$
$\boldsymbol{v}_{R+1}$
$\boldsymbol{u}_{R+1}$
$\boldsymbol{u}_M$
null space: Null($\boldsymbol{A}$)
dim: $N-R$
left null space: Null($\boldsymbol{A}^{\mathrm{H}}$)
dim: $M-R$
$\boldsymbol{\Sigma}^{+} = \begin{bmatrix} 1/\sigma_1 & 0 & & 0 & 0 \\ 0 & \ddots & & \vdots & \vdots \\ & & 1/\sigma_R & & \\ 0 & \cdots & & 0 & 0 \end{bmatrix}$